EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mock Final Examination

Course: CSE251 Electronic Circuits, Section-X
Instructor: Rashedul Amin Tuhin, Senior Lecturer, CSE Department
Full Marks: 30 (30 will be counted for final grading)
Duration: 90 minutes

Notes: There are **SIX** questions, answer ALL of them. Course Outcome (CO), Cognitive Level, Mark, for each question are mentioned at the right margin.

Q1. A 741-op amp has an open-loop voltage gain of 3×10^5, input resistance of $3\ M\Omega$, and output resistance of $60\ \Omega$. The op amp is used in the circuit of *Figure 1*. **Find** the closed loop gain $\frac{v_o}{v_i}$. **[CO3, C3, Mark: 04]**
Using the op-amp equivalent circuit model, **determine** current $\boldsymbol{i}$ when $v_s = 3\ V$.

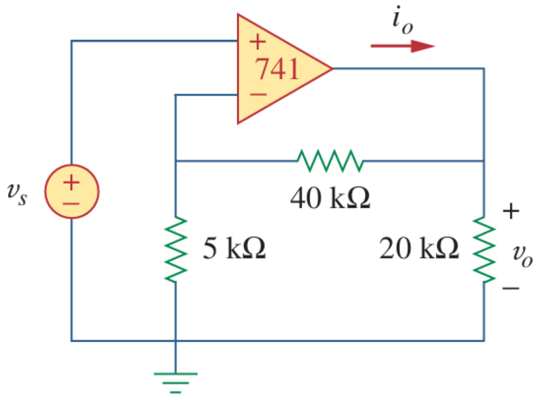

Figure 1

Q2. **Design** a circuit using a series of op-amps to generate the following output, **[CO3, C4, Mark: 09]**

$$v_o(t) = -\int_0^t (2v_1 - 3v_2 - 3v_3)\, dt$$

If the integrating capacitor is $C = 1\mu F$, **obtain** other component values.

Q3. For an *n-channel enhancement type* MOSFET, **[CO1, C2, Mark: 04]**

a) **Draw** the structure and mark the components.

b) **Describe** the MOSFET operation when,
   i. $V_{GS} = 0$ V, $V_{DS} = 0$ V
   ii. $V_{GS} > V_T$, $V_{DS} = 0$ V
   iii. $V_{GS} > V_T$, $V_{DS} > 0$ V

c) **Explain** with diagrams how the MOSFET can act as an amplifier.

Q4. A *0.18-μm* fabrication process is specified to have an *oxide thickness* of 4nm, *electron mobility* of 450cm^2/V·s, and *threshold voltage of 0.5V*. **Find** the value of the *process transconductance parameter* $k_{n'}$. **[CO1, C3, Mark: 03]**

For a MOSFET with *minimum length* fabricated in this process, **find** the required *width* so that the device exhibits a *channel resistance* of 1000Ω when V_{GS} = 2V.

Q5. For a *0.18-μm* fabrication process for which the *oxide thickness* is given as 4nm, *electron mobility* of 450cm^2/V·s, **find** the *oxide capacitance per unit area* and the *process transconductance parameter* $k_{n'}$. **[CO2, C3, Mark: 03]**

Find the *overdrive voltage* required to operate the MOSFET in the saturation region, where the *aspect ratio* is 20 and *saturation current* is 0.3m A.

Q6. For the circuit given in Figure 2, **determine** *kn*, I_{DQ}, V_{GSQ} and V_{DS}. **[CO2, C4, Mark: 07]**

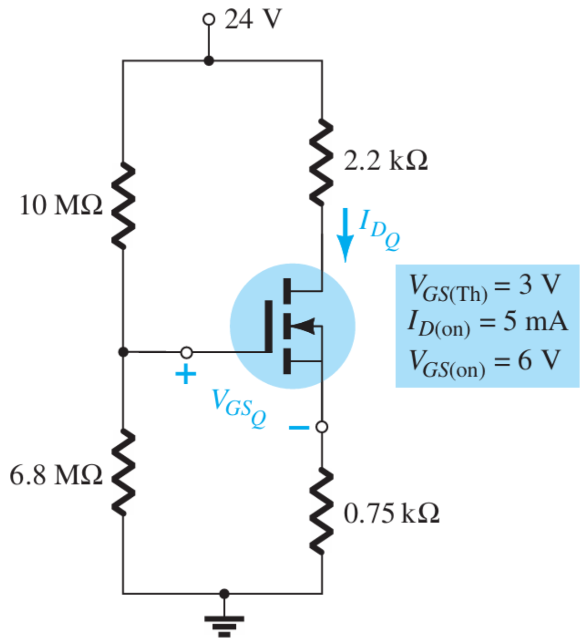

Figure 2